

Week 01. Part 1. Introduction

Population and Sample. Description and Inference.

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Content

- The Population
- The Sampling
- Population and sampling: summary
- How to conclude about population using random samples and statics?
- Homework: Example of statistical inference in a research
- Takeaways checklist

The Population

- See: Ch. 1.3.3 Sampling from a population in ([Diez, Cetinkaya-Rundel, and Barr 2022](#))

The Population

- “... is a natural, geographical, or political collection of people, animals, plants, or objects”
- “... is the **physical collection**, ... from which we derive a set of values or the variable of interest”
- specific to our **research question** and **the research design**
- defines the scope of our study, **its external validity**.

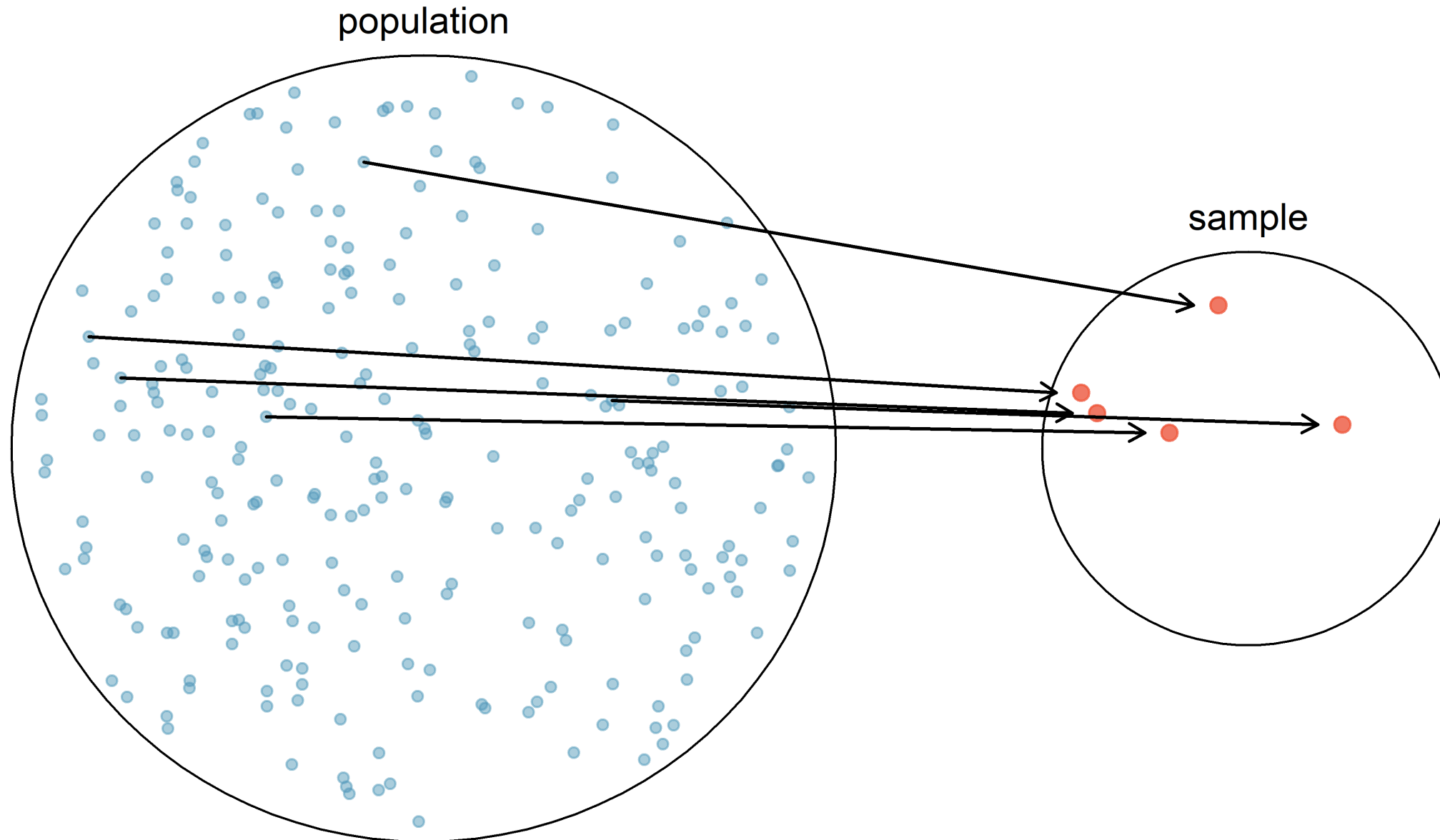
It is generally impossible to have data for the entire population

Source: ([Dowdy, Weardon, and Chilko 2004](#)).

The Sampling

- See also book “Sampling” by ([Thompson 2012](#))
- The World Bank. Development Research in Practice (DIME)
<https://dimewiki.worldbank.org/Sampling>

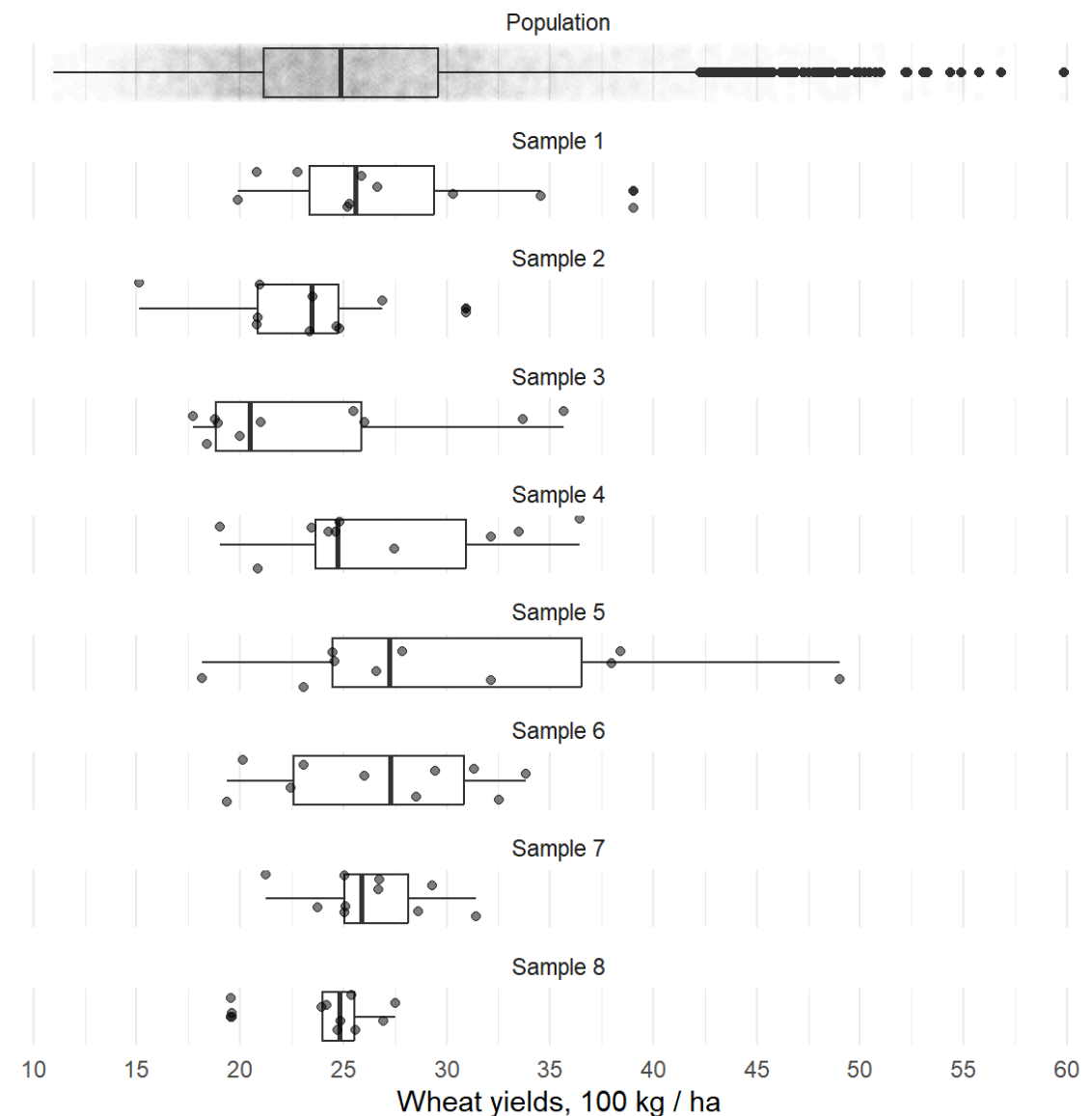
Simple Random Sample (1)



Source: [OpenIntroStat/openintro-statistics](https://openintrostat.github.io/openintro-statistics/), (Diez, Cetinkaya-Rundel, and Barr 2022)

The Simple Random Sample (2)

- **Simple sample** is a random subset of observations from a population.
 - **Random** means that every observation from the population has an **equal probability of being sampled**.
- Recall: What is the average wheat yield in 100 kg per ha?
- How different our samples are going to be?

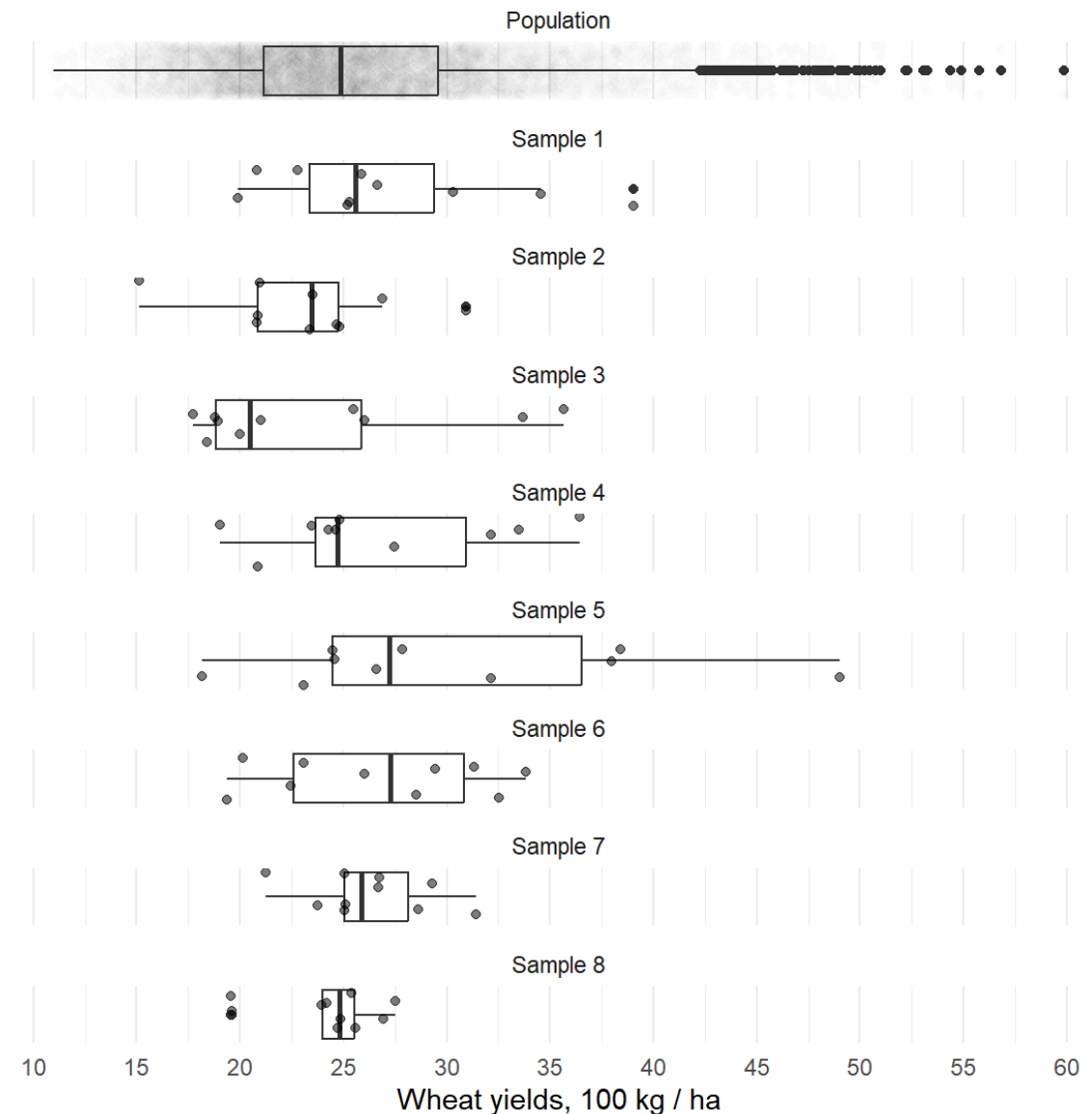


The Simple Random Sample (3)

Each sample results with a different estimate of the yields:

Type	Yields
Sample 1	27.04
Sample 2	23.19
Sample 3	23.57
Sample 4	26.66
Sample 5	30.23
Sample 6	26.68
Sample 7	26.30
Sample 8	24.23

- What is “**the correct**” average yields?
- Which sample estimates “**the correct**” average yield?
 - How to pick one?
 - Can we pick one?



The power of a random sample

If every sample is unique, why random sample is so important?

A random sample can approximate the population

- **when its size (N) increases to infinity!**

This is the **Law of Large Numbers (LLN)**

LLN makes random samples a powerful tool for understanding the population.

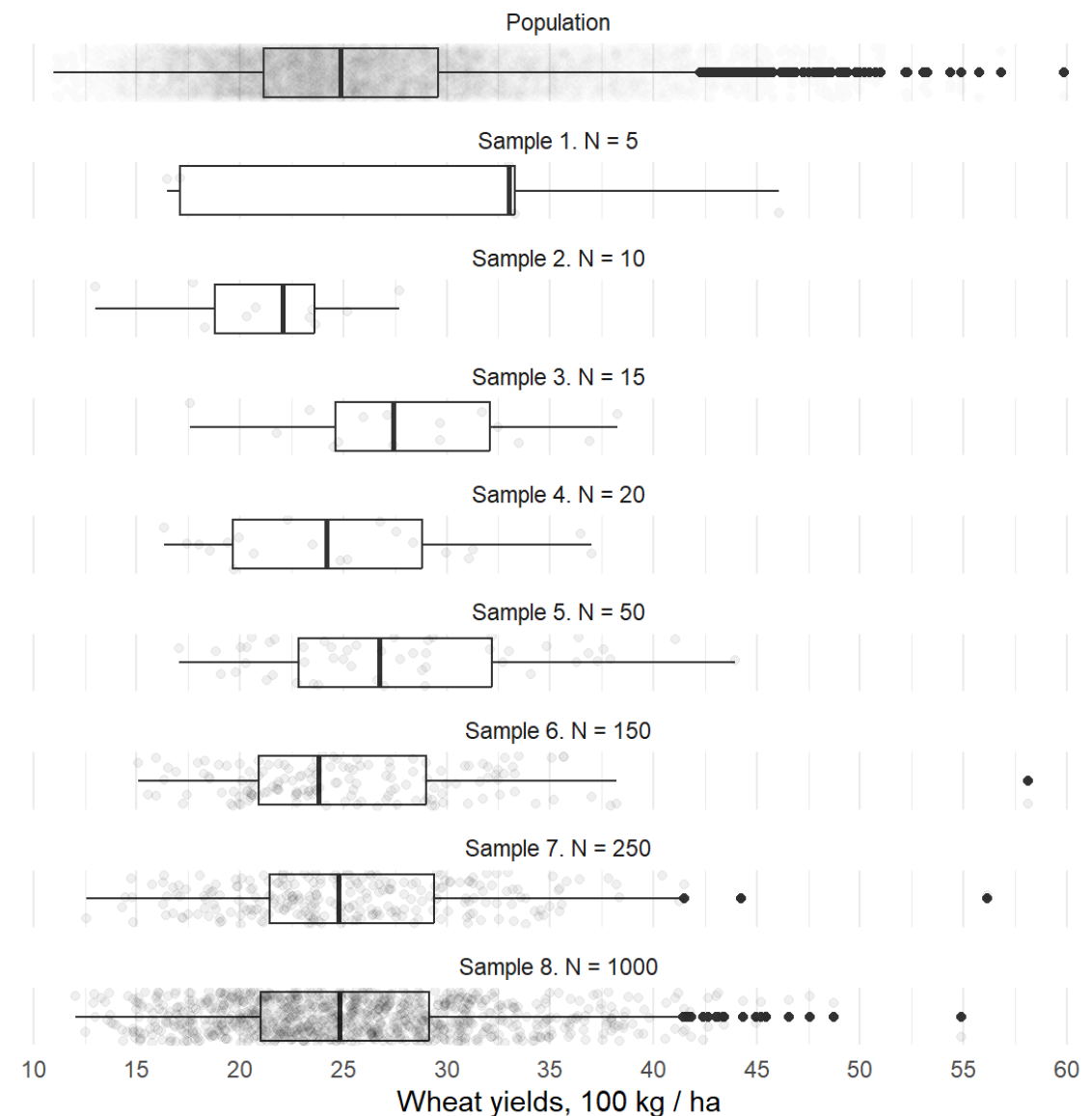
Law of Large Numbers: action

Let us gradually increase the sample size!

- What are the differences between samples?

Type2	Yields
Sample 1. N = 5	29.19
Sample 2. N = 10	21.36
Sample 3. N = 15	28.31
Sample 4. N = 20	24.72
Sample 5. N = 50	27.54
Sample 6. N = 150	25.22
Sample 7. N = 250	25.44
Sample 8. N = 1000	25.48

With the increasing size, the sample becomes more similar to the population!



Law of Large Numbers: limitation

Sampling is costly: Larger sample size = larger expenses.

Each sample is still a random draw!

- Can we be sure that our sample estimates the right average?
- If a population has minorities, how accurately can a random sample represent them?

Solutions:

- Apply appropriate sampling techniques: stratified and clustered sampling.
- Embrace uncertainty and use inferential statistics.

Simple random sample limitations (1)

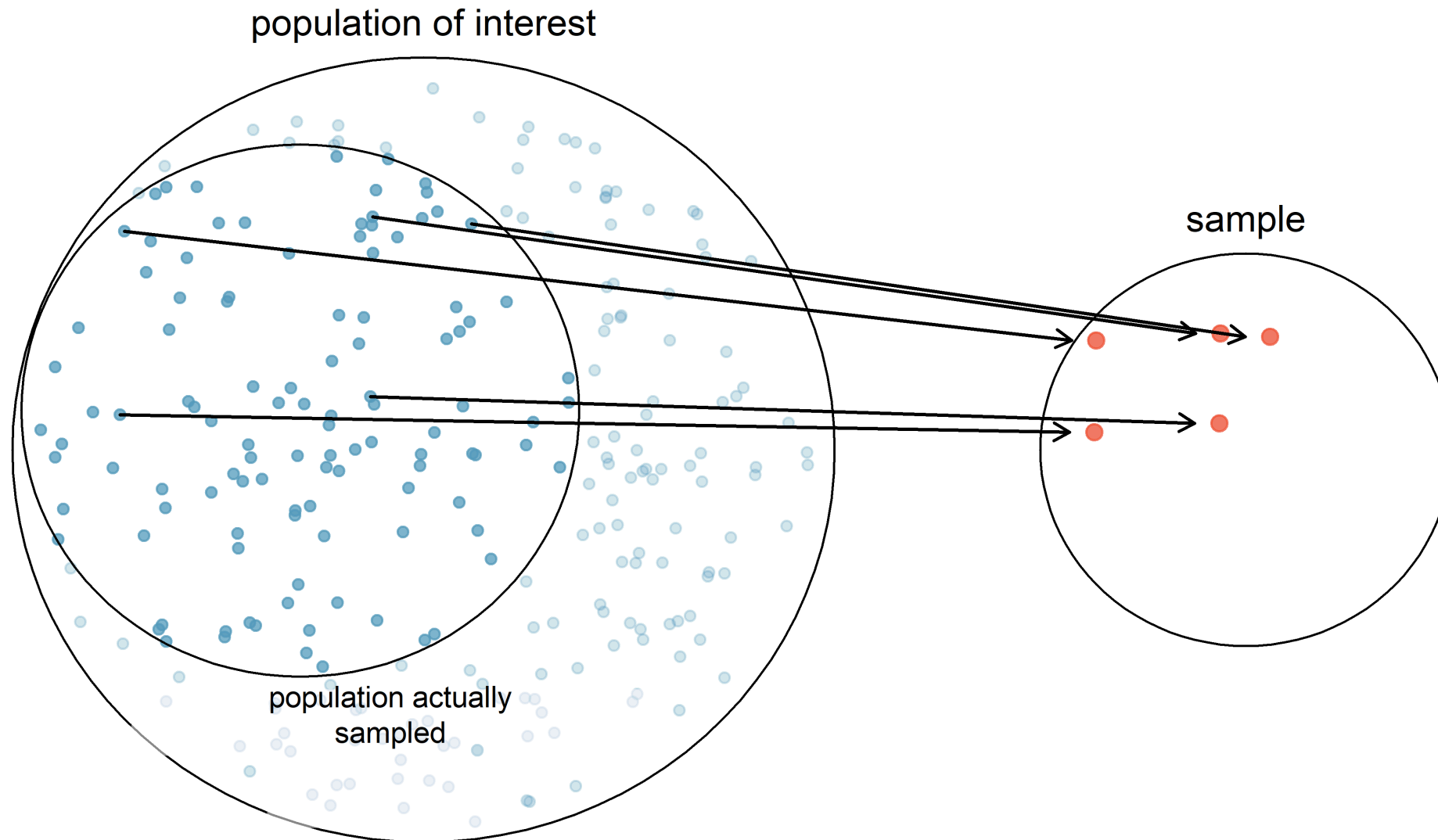
Desired sample



Source: [OpenIntroStat/openintro-statistics](https://openintrostat.github.io/openintro-statistics/) (Diez, Cetinkaya-Rundel, and Barr 2022)

Simple random sample limitations (2)

Actual sample



Source: [OpenIntroStat/openintro-statistics](https://openintrostat.github.io/openintro-statistics/) (Diez, Cetinkaya-Rundel, and Barr 2022)

Simple random sample limitations (3)

Key problems:

- Underlining structures in the population: **strata / clusters**.
- No-response
- Convenience-sample
 - where individuals who are easily accessible are more likely to be included in the sample.
- Under such limitations our sample is not **representative** or contains **measurement errors**.

Solutions:

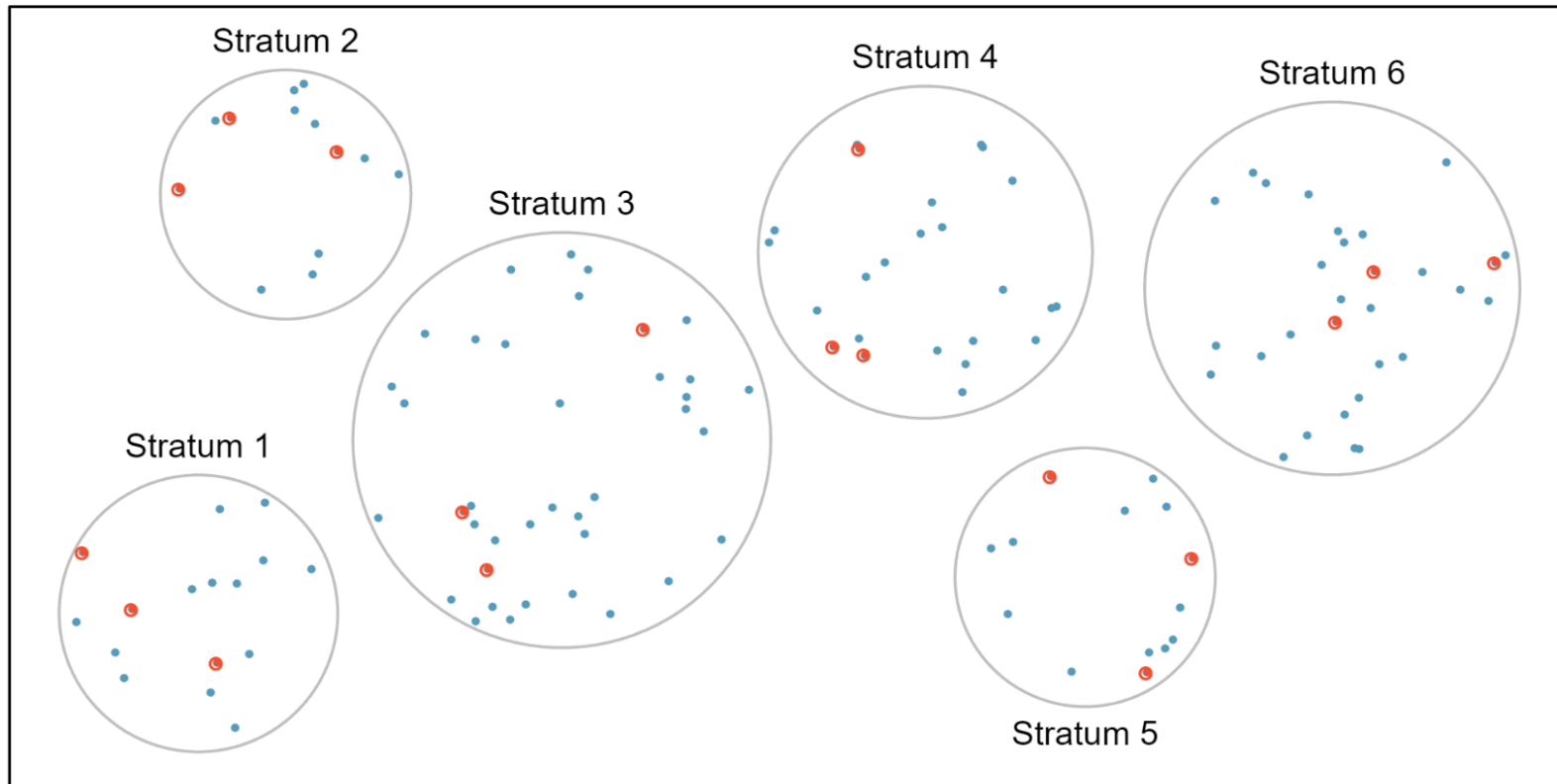
Advanced sampling techniques

Those exist to balance between data quality and data collection costs.

- Simple, Stratified, Cluster, and Multistage sampling;
- Population-weighted sampling;

To learn more, see Sampling by ([Thompson 2012](#))

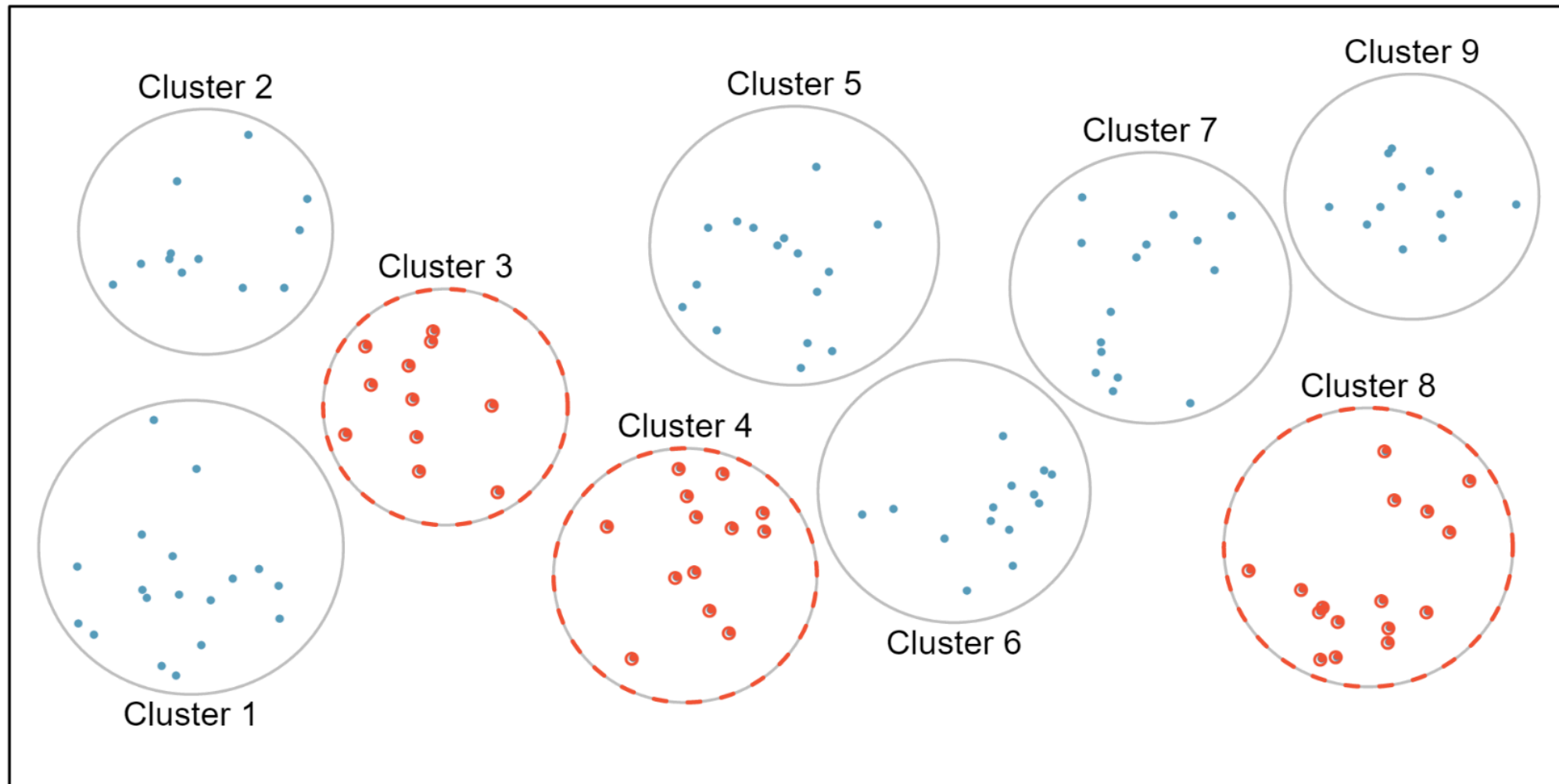
Stratified sampling



The population is divided into groups **strata**.

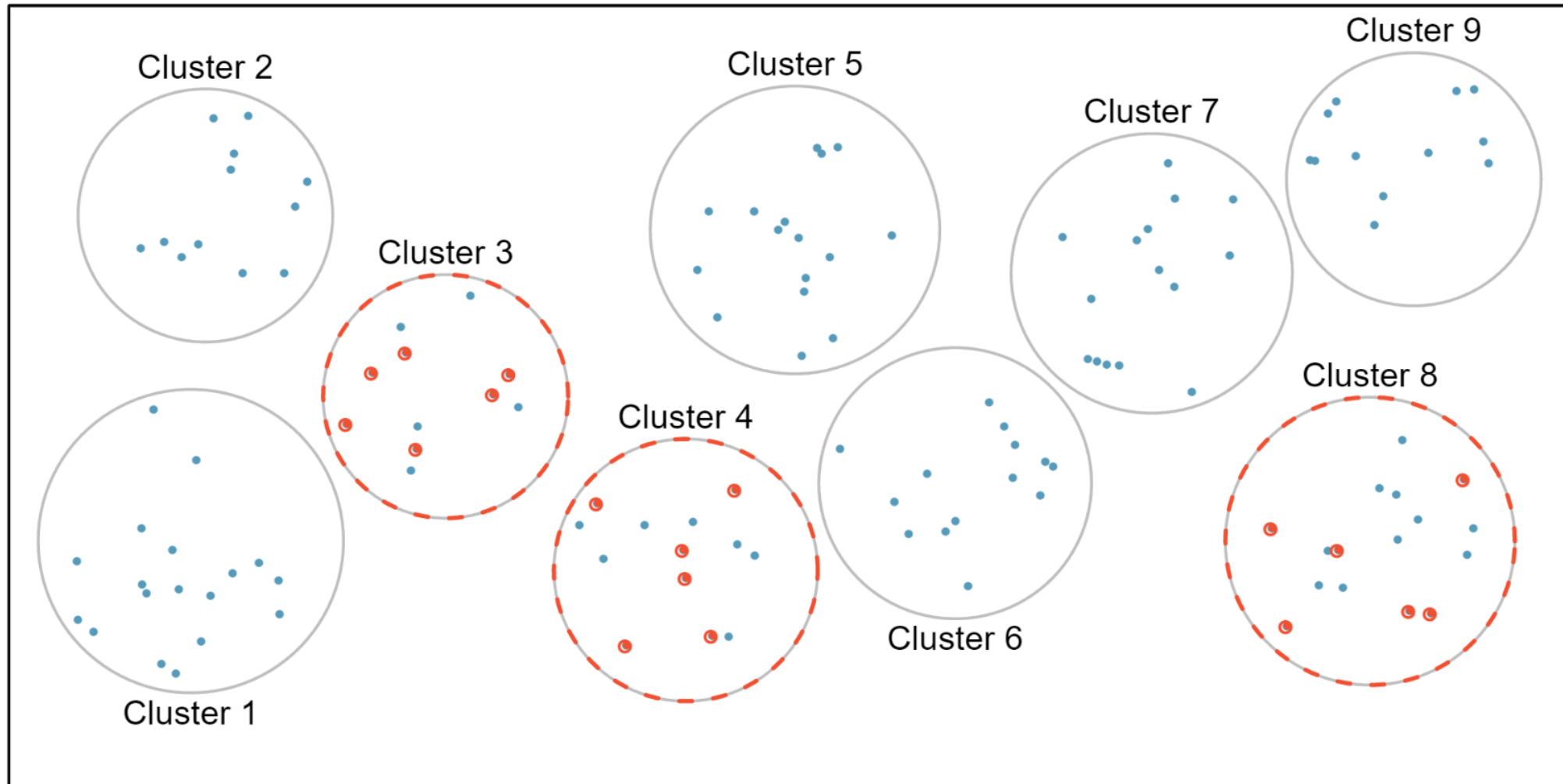
- strata combines **similar cases** especially with respect to response variables;
- within-strata: simple random sampling

Cluster sampling



- Population is broken into clusters (by location usually).
- We randomly select several clusters and sample them completely.

Multistage sampling



- Cluster + satisfied sampling
- Cluster + simple sampling

Population and sampling: summary

- Research objectives are to conclude about the population.
- But the population is **never available**, instead we have **RANDOM samples**.
- Random samples are all different: result with different estimates.
 - Thus, descriptive statistics based on each sample is not identical to one based on the population.

Population and samples by research question

Discuss what are the (1) populations; (2) variables of interest; (3) data source; (4) sample based on the exemplary research questions below:

1. What is the average poverty headcount rate in the world/a country?

- Population: people or households;
- Sample examples: household surveys.
- Data source: The Living Standards Measurement Study (LSMS)
<https://www.worldbank.org/en/programs/lsms>

At home: discuss the same for other research questions mentioned earlier.

- What is the average wheat yield in Hessen?
- How does access to education impact income inequality?
- What are the factors influencing malnutrition rates among children in a specific region?
- How does access to clean water and sanitation facilities impact child mortality rates?
- What are the barriers to accessing healthcare services among marginalized communities?

How to conclude about population using random samples and statics?

- Any ideas?
- There are:
 - Descriptive statistics
 - Inferential statistics

Descriptive statistics

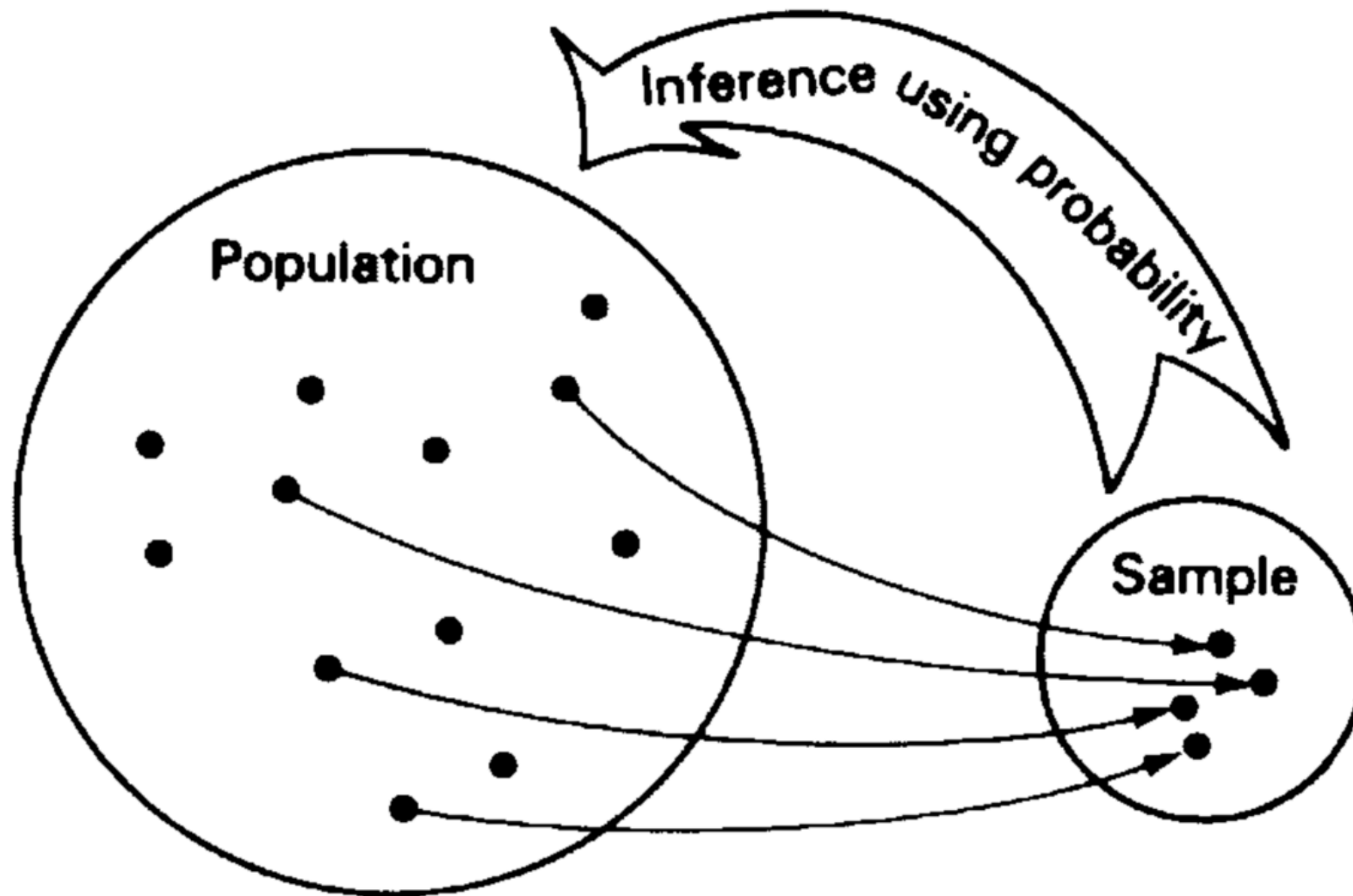
- AKA Data Analysis:
 - uses statistical methods to described observed data (data that comes from samples)
- But because each sample has a random component and is different from the population,
 - **descriptive statistics** does not tell us much about the **population**...

Inferential statistics

- Answers questions **about the population** using:
 - random **samples**, and the **probability theory**
- Does so by:
 - Establishing **statistical hypothesis**;
 - Using **statistical methods** and **probability theory** to **reject or accept** it.

The process of statistical inference

Source: (Dowdy, Weardon, and Chilko 2004)



Statistical inference is based on **statistical hypothesis testing (HT)**.

Statistical hypothesis: basics (1)

As we rely on a samples, we call any measure (mean, median, variance) computed based on such samples as:

- **the estimates of {mean, median, variance}**. We denote estimates with a “hat” “ $\hat{}$ ”, for example: $\{\hat{\text{mean}}, \hat{\text{median}}, \hat{\text{variance}}\}$

Based on a sample we can **only calculate an estimate** of a $\hat{\text{mean}}$, $\hat{\text{median}}$, $\hat{\text{variance}}$

Statistical hypothesis: basics (2)

To infer that the true {mean, median, variance} in the population is not different from the estimate of $\{\widehat{\text{mean}}, \widehat{\text{median}}, \widehat{\text{variance}}\}$, we:

1. formulate two statistical hypothesis:

- **H0** (the null hypothesis):
 - “The estimate of {...} is equal to ZERO in the population”.
- **H1** (the alternative hypothesis):
 - “The estimate of {...} is NOT equal to zero in the population”.

Statistical hypothesis: basics (3)

2. use **statistical tests** and the **probability theory** to reject **H0** (or fail to reject H0) and conclude either:

- We reject **H0** in favour of **H1**:
 - with the probability P (p-value) that we've made a mistake by rejecting H0 (Type I error).
- We accept **H0** (fail to reject H0)
 - because available data is insufficient to reject it;
 - without implying that H0 is the right hypothesis (Type II error);

Goals of the Part 1: Quantitative Research Methods

- Grasp the difference between description and inference.
- Learn how to:
 - apply basic methods of descriptive statistics;
 - perform statistical inference;
 - make conclusions based on the results of statistical inference;
 - communicate our story using statistical tools;

Homework: Example of statistical inference in a research

Read through the examples

1. Research question
2. Research design
3. Data collection
4. Data analysis

Research question

Do extension services in agriculture improve rice yields in “a province” of Vietnam?

- **The population: all rice producers in “a province” of Vietnam.**
- **The variables: rice yields in tons/ha and amount and quality of extension services received.**

Research Design (simplified)

Randomized Control Trial (RCT) experiment:

- Divide population into **clusters** and **strata**.
- **Clusters** are similar in structure but are located in different places: **villages**
 - Treatment group: 4 clusters with extension services.
 - Control group: 4 clusters without such.
- Same **strata** are present in each cluster: **small, medium, and large farmers**.
 - **Stratified Simple Random Sampling of 200 farmers in each cluster proportionally representing all strata.**
- Final sample size 1600 observations.

Data Collection: Two waves

Wave 1: pre-intervention

- Exist to establish the baseline;
- Establish the **check and balance tables**
 - To ensure that **on average, treatment and control groups are similar.**

Wave 2: post-intervention

- Exist to check the hypotheses.
- Collects data on the variables of interest.

Data analysis

1 year after the treatment, we repeat data collection and measure rice yields:

Treatment group

Response rate: 1483

Rice yields: 6.0 tone per ha

Control group

Response rate: 1542

Rice yields: 5.8 tone per ha

Can we conclude extension services have a positive effect on the rice yields in Vietnam?

Let us see what rice yields looks like in the world:

<https://ourworldindata.org/grapher/rice-yields>

To conclude about the effect of intervention, we need to

- “embrace” the uncertainty!
- perform statistical inference,
- consider **Variance** not only point-estimates.

Takeaways checklist

Homework

! Important

Study all prerequisites on your own: see Ilias.

- Familiarize with the World Bank DIME initiative
<https://www.worldbank.org/en/research/dime>
 - And Dime knowledge base:
<https://www.worldbank.org/en/research/dime>
 - Find one project on their website and try to understand what type of data collection was used there and how it was implemented.

Key concepts covered in the lecture

- The population and a sample;
- Simple Random Sample: (non-) representative sample, convenience sample, measurement error.
- Other sampling technique: stratified, clustered and multistage sampling.
- Descriptive statistics.
- Inferential statistics.
- Exemplary research design.

Example of the exam questions

- Why do we need samples and random sampling?
- Why random sampling is such a powerful tool?
- When does random sampling fail?
- What can descriptive statistics tell us about the population?
- Why do we need inferential statistics?
- When does simple random sampling fail?
- What key random sampling techniques exist and why do we need them?

References

- Diez, David, Mine Cetinkaya-Rundel, and Christopher D Barr. 2022. *OpenIntro Statistics: Fourth Edition*. OpenIntro. <https://www.openintro.org/book/os/>.
- Dowdy, Shirley, Stanley Weardon, and Daniel Chilko. 2004. *Statistics for Research*. John Wiley & Sons, Inc. <https://doi.org/10.1002/0471477435>.
- Thompson, Steven K. 2012. *Sampling*. John Wiley & Sons, Inc. <https://doi.org/10.1002/9781118162934>.